attempting to do so is a drug addict, mental patient or charlatan. Consequently, though we have a detailed map of the mental landscape of Harvard psychology students, we know far less about the mental landscapes of Native American shamans, Buddhist monks or Sufi mystics.[2]

And that is just the Sapiens mind. Fifty thousand years ago, we shared this planet with our Neanderthal cousins. They didn't launch spaceships, build pyramids or establish empires. They obviously had very different mental abilities, and lacked many of our talents. Nevertheless, they had bigger brains than us Sapiens. What exactly did they do with all those neurons? We have absolutely no idea. But they might well have had many mental states that no Sapiens had ever experienced.

Yet even if we take into account all human species that ever existed, that would still not exhaust the mental spectrum. Other animals probably have experiences that we humans can barely imagine. Bats, for example, experience the world through echolocation. They emit a very rapid stream of high-frequency calls, well beyond the range of the human ear. They then detect and interpret the returning echoes to build a picture of the world. That picture is so detailed and accurate that the bats can fly quickly between trees and buildings, chase and capture moths and mosquitoes, and all the time evade owls and other predators.

The bats live in a world of echoes. Just as in the human world every object has a characteristic shape and colour, so in the bat world every object has its echo-pattern. A bat can tell the difference between a tasty moth species and a poisonous moth species by the different echoes returning from their slender wings. Some edible moth species try to protect themselves by evolving an echo-pattern similar to that of a poisonous species. Other moths have evolved an even more remarkable ability to deflect the waves of the bat radar, so that like stealth bombers they fly around without the bat knowing they are there. The world of echolocation is as complex and stormy as our familiar world of sound and sight, but we are completely oblivious to it.

One of the most important articles about the philosophy of mind is titled 'What Is It Like to Be a Bat?'[3] In this 1974 article,